

CareFlow AI

Product AI Documentation

Runtime AI Architecture, Workflows, Safety and Observability

Purpose

This document describes how artificial intelligence is integrated into the CareFlow AI product to support post-discharge patient follow-up, clinical triage, escalation assessment, and structured documentation.

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1 Product Overview

CareFlow AI is a healthcare operations platform designed to coordinate post-discharge patient follow-up.

The product connects:

- hospitals and their staff;
- patient discharge information;
- outreach campaigns;
- eligibility processing;
- an operational outreach queue;
- patient responses;
- AI-assisted response analysis;
- clinical escalation;
- human review;
- controlled healthcare/EHR operations;
- notifications;
- audit and observability.

The AI functionality is therefore one part of a larger healthcare operations workflow rather than an isolated chatbot.

2 Role of AI in the Product

The product AI layer supports four primary logical responsibilities:

1. **Voice / Response Intake**
2. **Clinical Triage**
3. **Escalation Decision Support**
4. **Documentation**

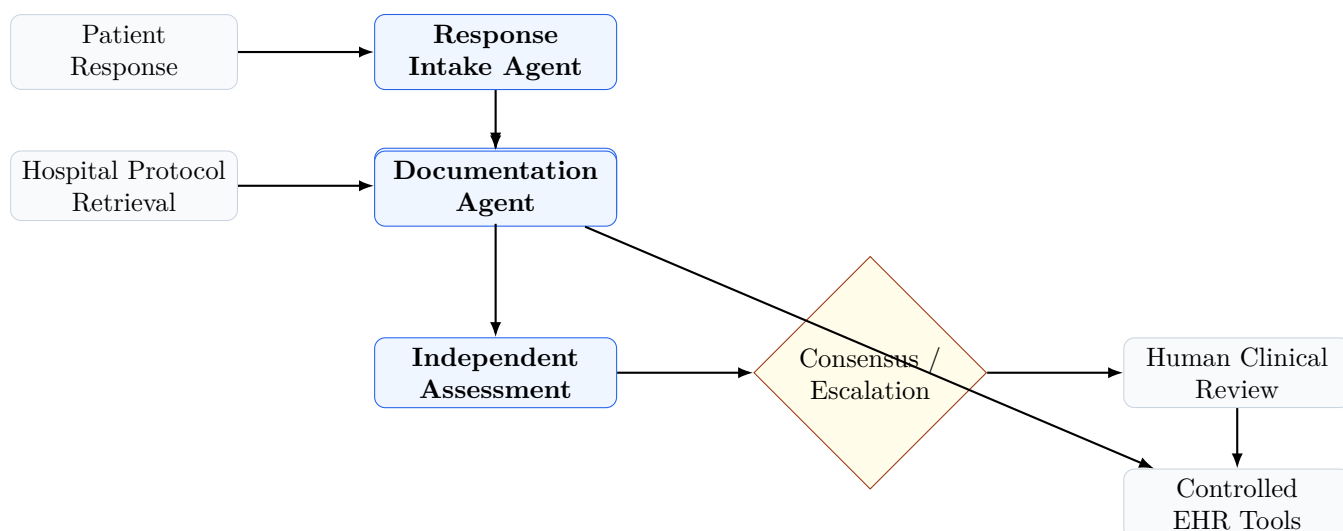
The AI does not independently control the complete application workflow. Important actions remain controlled by backend services and, where appropriate, human clinical reviewers.

Core Product Principle

AI produces structured assessments and recommendations. The application backend validates those outputs and controls downstream state transitions and authorized actions.

3 Product AI Architecture

The following conceptual architecture shows how the AI layer interacts with the rest of the product.



4 AI Input and Context

The AI processing pipeline may use the following categories of information:

- patient follow-up response;
- structured encounter information;
- discharge information;
- relevant patient context;
- hospital-specific protocols;
- retrieved knowledge;
- previous workflow state where required.

Only information required for the specific AI task should be provided to the corresponding component.

Patient-provided text and retrieved documents are treated as **untrusted content** and must not be allowed to override system instructions, authorization rules, or application safety controls.

5 AI Component 1 – Response Intake

The Response Intake component converts the patient’s follow-up response into structured information suitable for downstream processing.

The input may originate from:

- typed text;
- recorded audio;
- supported multilingual patient communication.

The intake stage focuses on understanding the response rather than making the final clinical decision.

5.1 Responsibilities

- identify the patient's reported concerns;
- extract relevant symptoms or indicators;
- preserve important patient statements;
- normalize the response into structured data;
- identify uncertainty or missing information.

5.2 Boundary

The intake component should not independently modify EHR records, change patient authorization, or directly trigger arbitrary tools.

6 AI Component 2 – Clinical Triage

The Clinical Triage component analyzes the structured patient response using relevant hospital protocol information.

The target classification is:

Routine | Concerning | Urgent | Uncertain

The classification is accompanied by supporting information rather than being treated as a standalone answer.

6.1 Expected Triage Output

A representative structured result is:

```
{
  "classification": "urgent",
  "observedIndicators": [
    "severe chest pain",
    "difficulty breathing"
  ],
  "evidence": [
    "Patient reports worsening chest pain",
    "Patient reports difficulty breathing"
  ],
  "protocolReferences": [
    "hospital_protocol_reference"
  ],
  "confidence": 0.XX,
  "uncertainty": [],
  "escalationRecommendation": "human_review"
}
```

The exact numerical confidence and provider-specific behavior depend on the runtime AI implementation and configuration.

7 Protocol-Grounded AI

AI responses should be grounded in hospital-specific clinical protocols where the relevant protocol information is available.

The conceptual retrieval workflow is:

1. identify the hospital/tenant;
2. retrieve relevant protocol information;
3. provide the retrieved information as controlled context;
4. ask the AI to reason only within the supplied context and system instructions;
5. include protocol references in the structured result;
6. validate the result before downstream use.

This reduces dependence on unsupported general medical knowledge.

Grounding Rule

Retrieved protocol documents provide contextual evidence. They do not have authority to override system instructions, authorization policies, backend controls, or safety constraints.

8 Multiple Independent Assessments

For escalation-sensitive cases, the system is designed around multiple independent assessments rather than relying on a single AI assessment.

Conceptually:

$$A_1 = f_1(\text{Response}, \text{Protocol})$$

$$A_2 = f_2(\text{Response}, \text{Protocol})$$

The outputs are then compared by an escalation/consensus component.

$$\text{Decision} = \text{Consensus}(A_1, A_2, \text{Protocol})$$

The assessments may use different independent reasoning paths or equivalent independent evaluation mechanisms.

The purpose is to identify:

- agreement;
- disagreement;
- missing evidence;
- uncertainty;
- potentially unsafe conclusions.

9 Disagreement Handling

Disagreement between assessments is treated as an explicit signal.

For example:

Assessment A	Assessment B	System State
Routine	Routine	Continue workflow
Concerning	Concerning	Continue with configured handling
Urgent	Urgent	Human review / escalation
Routine	Urgent	Disagreement → review
Concerning	Uncertain	Uncertainty → review

The system is designed to avoid silently treating conflicting assessments as agreement.

When the available evidence does not support a confident classification, the system can preserve uncertainty and route the case for human review.

10 Escalation Decision System

The escalation layer combines the available AI assessments and workflow rules.

Its responsibilities include:

- evaluating assessment agreement;
- identifying urgent classifications;
- identifying uncertainty;
- detecting disagreement;
- determining whether human review is required;
- producing a structured escalation decision.

The escalation layer does not replace clinical staff.

10.1 Urgent Case Example

A patient may submit:

“I am having severe chest pain and difficulty breathing since morning. The pain is getting worse.”

The product workflow is:

1. Patient response received.
2. Response is processed.
3. Relevant hospital protocol is retrieved.
4. Independent AI assessments are generated.
5. Assessments are compared.
6. Urgent indicators are identified.
7. Escalation is created.
8. Case enters human clinical review.

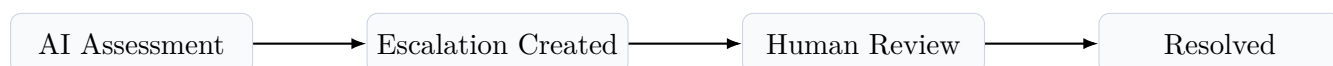
9. Staff member reviews the case.
10. Controlled downstream actions can be performed.
11. The workflow is audited.

The important product behavior is that an urgent AI assessment results in a controlled escalation workflow rather than an autonomous clinical action.

11 Human-in-the-Loop Workflow

Human review is an explicit part of the product.

A simplified state flow is:



The reviewer can inspect the relevant case information and make the appropriate operational decision within the application's authorized workflow.

12 Documentation Agent

The Documentation Agent converts structured workflow information into consistent documentation.

Potential inputs include:

- patient response;
- triage classification;
- observed indicators;
- protocol references;
- escalation outcome;
- human review information.

Documentation generation should remain grounded in the information available in the workflow.

The Documentation Agent must not invent clinical observations, diagnoses, actions, or patient information.

13 Controlled AI Tools

AI components should not receive unrestricted application access.

Tool access is controlled by backend services.

Examples include:

- retrieving authorized hospital protocols;
- reading permitted patient context;
- creating an escalation through a controlled backend operation;
- generating documentation;

- updating permitted workflow state;
- creating controlled EHR records where supported.

The backend remains responsible for:

- authentication;
- authorization;
- tenant isolation;
- input validation;
- state transitions;
- audit logging.

14 EHR Integration Boundary

The AI layer does not directly manipulate the underlying healthcare database.

Instead, EHR operations are exposed through controlled application interfaces.



This boundary prevents the language model from obtaining unrestricted database access.

15 AI Prompt Design

Runtime AI prompts are logically separated by responsibility.

Prompt		Purpose	Expected Output
Response Intake		Understand and structure patient response	Structured response information
Clinical Triage		Classify patient-reported indicators using supplied context	Triage JSON
Escalation	Assessment	Assess whether escalation or human review is required	Escalation recommendation
Consensus		Compare independent assessments	Consensus / disagreement result
Documentation		Create structured workflow documentation	Validated documentation
Evaluation		Assess AI output against expected behavior	Evaluation result

16 Runtime Prompt Structure

A conceptual runtime prompt contains separate instruction layers.

```

SYSTEM INSTRUCTIONS
|
+-- Role
+-- Safety boundaries
+-- Output schema
  
```

```
    +-- Tool authorization rules
    |
CONTEXT
    |
    +-- Hospital / tenant context
    +-- Relevant patient workflow context
    +-- Retrieved protocol information
    |
PATIENT CONTENT
    |
    +-- Patient response
    |
TASK
    |
    +-- Required classification / assessment
    |
OUTPUT
    |
    +-- Structured machine-readable result
```

Patient content is treated as data.

It must not be interpreted as a higher-priority instruction.

17 Structured Output Validation

AI output is not accepted blindly.

The backend should validate:

- required fields;
- allowed classification values;
- data types;
- protocol references;
- escalation recommendation;
- uncertainty representation;
- consistency with workflow state.

A simplified validation pipeline is:

AI Output → Schema Validation → Business Rules → Workflow Transition

Invalid or incomplete output should enter an explicit failure or manual-review path instead of silently being treated as valid.

18 AI Safety Boundaries

The product follows several AI safety boundaries.

18.1 Patient Input Is Untrusted

Patient messages can contain arbitrary language, instructions, or claims. They must not override system instructions.

18.2 Retrieved Documents Are Untrusted

Retrieved protocol content is context, not executable instructions.

18.3 AI Cannot Bypass Authorization

The AI cannot independently grant itself access to patient records, hospital data, or protected tools.

18.4 AI Cannot Directly Perform Uncontrolled Clinical Actions

Critical operations remain behind backend-controlled interfaces.

18.5 Uncertainty Is Preserved

When evidence is insufficient, the system should be able to represent uncertainty rather than forcing a confident classification.

19 AI Observability

AI operations should be observable without unnecessarily logging sensitive healthcare information.

Useful AI telemetry includes:

- AI component / agent;
- model/provider identifier where available;
- purpose;
- request timestamp;
- latency;
- success/failure;
- structured validation status;
- token usage where available;
- estimated cost where available;
- correlation/request identifier.

Sensitive patient content should not be unnecessarily copied into general application logs.

20 AI Failure Handling

The AI layer must account for external provider and processing failures.

Potential failure modes include:

Failure	System Response	Operational Outcome
AI timeout	Timeout and controlled retry	Retry / failure state

AI provider failure	Retry with bounded attempts	Manual recovery if exhausted
Malformed AI output	Schema validation failure	Do not apply result
Missing protocol	Explicit uncertainty	Review / configured fallback
Assessment disagreement	Consensus failure / uncertainty	Human review
Documentation failure	Retry / explicit failure state	Manual recovery if required
Tool failure	Controlled error	Do not perform unauthorized action

The application should never interpret an infrastructure failure as a successful clinical assessment.

21 AI Usage in the Patient Follow-Up Workflow

The complete product workflow can be represented as:

1. Hospital creates or runs a follow-up campaign.
2. Eligibility processing identifies patients.
3. Eligible patients enter the outreach queue.
4. Queue capacity controls outreach concurrency.
5. Patient receives outreach.
6. Patient responds using text or supported audio.
7. Response is captured.
8. AI processes the response.
9. Relevant protocol context is retrieved.
10. Clinical assessments are generated.
11. Assessments are compared.
12. Routine cases continue through the configured workflow.
13. Concerning or uncertain cases follow configured handling.
14. Urgent or disagreement cases can create human review.
15. Reviewer handles the escalation.
16. Controlled documentation/EHR operations can occur.
17. Audit events record important workflow transitions.

22 AI and Human Responsibilities

AI Responsibilities	Human / Backend Responsibilities
Understand patient responses	Authorize application actions
Extract relevant indicators	Control workflow state
Perform structured triage assessment	Enforce business rules
Retrieve and use supplied protocol context	Manage tenant access

Identify uncertainty	Review escalated cases
Compare independent assessments	Make operational/clinical review decisions
Generate structured documentation	Approve and control EHR operations
Provide machine-readable recommendations	Maintain auditability and accountability

23 AI Evaluation

Evaluation should consider both model output quality and workflow safety.

Important evaluation dimensions include:

- classification correctness;
- urgent-case sensitivity;
- false-negative rate;
- false-positive rate;
- uncertainty handling;
- protocol grounding;
- evidence extraction;
- escalation recommendation;
- structured output validity;
- disagreement detection.

For an evaluation dataset:

$$\text{False Negative Rate} = \frac{\text{False Negatives}}{\text{False Negatives} + \text{True Positives}}$$

For safety-sensitive triage, false-negative behavior should receive specific attention because incorrectly failing to identify a case that requires escalation can have significant consequences.

24 Synthetic Demonstration Environment

The demonstration environment uses deterministic synthetic patient data.

The seeded demonstration population contains:

- 30 synthetic patients;
- 24 patients associated with Apollo Demo Hospital;
- 6 patients associated with CityCare Demo Hospital;
- mixed risk scenarios;
- deterministic outreach outcomes;
- retry scenarios;
- callback scenarios;

- escalation scenarios.

The same seeded demo population is reused by the queue simulation rather than creating a second unrelated patient population.

This allows the product workflow to be demonstrated consistently without using real patient information.

25 Demo AI Workflow

A representative demonstration focuses on one urgent case.

25.1 Step 1 – Patient Response

The patient submits either text or recorded audio.

Example text:

“I am having severe chest pain and difficulty breathing since morning. The pain is getting worse and I am feeling very uncomfortable.”

25.2 Step 2 – AI Processing

The system:

1. receives the response;
2. processes the response;
3. retrieves relevant protocol context;
4. performs independent assessments;
5. compares the assessments;
6. identifies the escalation state.

25.3 Step 3 – Human Review

If the case is routed to human review:

1. escalation is created;
2. review task is displayed;
3. staff member acknowledges the case;
4. staff member reviews the available evidence;
5. the review is resolved;
6. controlled downstream actions may be performed;
7. audit information is recorded.

26 AI Auditability

Important AI-related operations should be traceable.

Examples include:

- AI assessment creation;
- AI assessment result;
- protocol retrieval;
- consensus decision;
- escalation creation;
- human review;
- reviewer override;
- documentation generation;
- controlled EHR updates.

The objective is to make it possible to understand:

What happened? Why did the workflow transition? Who or what performed the action?

27 Multi-Tenant AI Isolation

CareFlow AI supports multiple hospitals.

AI processing must therefore remain tenant-aware.

A simplified conceptual boundary is:

$$\textit{Hospital A} \rightarrow \textit{Hospital A Context} \rightarrow \textit{Hospital A AI Workflow}$$
$$\textit{Hospital B} \rightarrow \textit{Hospital B Context} \rightarrow \textit{Hospital B AI Workflow}$$

Protocol retrieval, patient context, campaign data, and operational records must respect hospital/tenant boundaries.

The AI layer should never be allowed to use another hospital's information simply because it is available elsewhere in the system.

28 Privacy and Data Minimization

The product is designed around minimizing unnecessary exposure of healthcare information.

Key principles include:

- use only information required for the task;
- avoid unnecessary sensitive information in logs;
- restrict data access through authorization;
- keep tenant boundaries enforced;
- avoid exposing raw database access to AI agents;

- maintain audit records for sensitive operations.

This prototype documentation does not claim regulatory compliance such as HIPAA or SOC 2 unless such compliance has been independently verified.

29 Known Limitations

The current product is a prototype and has limitations.

- AI output quality depends on the configured model/provider.
- Clinical classification is not a substitute for professional clinical judgment.
- Protocol coverage depends on the knowledge available to the application.
- Speech processing quality may vary by language, audio quality, accent, and provider.
- AI providers can experience latency, outages, or malformed responses.
- Synthetic demo data does not represent real-world patient population complexity.
- Production healthcare deployment would require additional security, privacy, compliance, clinical validation, monitoring, and operational controls.

30 Design Tradeoffs

Several deliberate tradeoffs are present in the product.

30.1 AI Assistance vs Deterministic Backend Logic

AI is used where language understanding is valuable. Deterministic backend code remains responsible for workflow state, authorization, queue capacity, and controlled actions.

30.2 Multiple Assessments vs Cost and Latency

Multiple assessments provide an additional mechanism for detecting disagreement but can increase AI usage, latency, and operational cost.

30.3 Protocol Grounding vs Coverage

Hospital-specific retrieval can improve contextual grounding, but the system becomes dependent on the availability and quality of the configured protocols.

30.4 Human Review vs Full Automation

Human review introduces additional operational work but provides an explicit control point for urgent, uncertain, or conflicting cases.

31 Product AI Principles

The CareFlow AI product follows these core principles:

1. **AI assists; it does not own the workflow.**
2. **Critical actions remain backend-controlled.**
3. **Patient content is treated as untrusted input.**

4. Retrieved knowledge is grounded context, not authority.
5. AI outputs are structured and validated.
6. Uncertainty is explicitly represented.
7. Disagreement is detected rather than hidden.
8. Urgent cases can be routed to human review.
9. AI activity is observable and auditable.
10. Synthetic data is used for demonstration.

32 Conclusion

AI in CareFlow AI is designed as a controlled component of a broader post-discharge healthcare operations platform.

The product uses AI to understand patient responses, perform structured clinical triage, compare independent assessments, identify cases that require escalation, and support documentation.

The system deliberately separates AI reasoning from authorization, workflow management, EHR access, and human clinical review.

This architecture allows the product to demonstrate the practical value of AI while maintaining explicit boundaries around safety, reliability, tenant isolation, auditability, and human oversight.